Started on	Friday, 5 December 2025, 9:17 PM
State	Finished
Completed on	Friday, 5 December 2025, 9:27 PM
Time taken	10 mins 38 secs
Marks	25.00/25.00
Grade	**100.00** out of 100.00

Question 1

Complete

Mark 1.00 out of 1.00

A foreign key must always reference:

- a. A primary key or unique key in another table
- b. A candidate key in the same table
- c. The first column of the parent table
- d. Any attribute in any table

Question 2

Complete

Mark 1.00 out of 1.00

A relationship that requires a linking (junction) table is usually:

- a. One-to-one
- b. One-to-many
- c. Recursive
- d. Many-to-many

Question 3

Complete

Mark 1.00 out of 1.00

If a transaction commits, the ability of the database to preserve changes even after a crash refers to:

- a. Isolation
- b. Durability
- c. Consistency
- d. Serializability

Question 4

Complete

Mark 1.00 out of 1.00

In a one-to-many relationship, the foreign key is typically stored in:

- a. A separate relationship table
- b. Both sides
- c. The "many" side
- d. The "one" side

Question 5

Complete

Mark 1.00 out of 1.00

Violation of which ACID property leads to dirty reads?

- a. Consistency
- b. Isolation
- c. Durability
- d. Atomicity

Question 6

Complete

Mark 1.00 out of 1.00

What is the output of COUNT(*) if a table has 5 rows containing NULL values?

- a. 0
- b. Only non-NULL rows counted
- c. 5
- d. NULL

Question 7

Complete

Mark 1.00 out of 1.00

What is the purpose of a unique key?

- a. To enforce uniqueness but allow a single NULL
- b. To enforce referential integrity
- c. To generate values automatically
- d. To allow duplicate values only once

Question 8

Complete

Mark 1.00 out of 1.00

Which ACID property ensures that a transaction's operations are treated as a single unit?

- a. Consistency
- b. Atomicity
- c. Durability
- d. Isolation

Question 9

Complete

Mark 1.00 out of 1.00

Which ACID property ensures that concurrently running transactions do not affect each other's intermediate states?

- a. Durability
- b. Consistency
- c. Isolation
- d. Atomicity

Question 10

Complete

Mark 1.00 out of 1.00

Which ACID property guarantees that a transaction moves the database from one valid state to another?

- a. Atomicity
- b. Durability
- c. Consistency
- d. Isolation

Question 11

Complete

Mark 1.00 out of 1.00

Which aggregate function counts only non-NULL values?

- a. COUNT(*)
- b. COUNT(column_name)
- c. AVG(column_name)
- d. SUM(column_name)

Question 12

Complete

Mark 1.00 out of 1.00

Which aggregate function ignores NULLs by default?

- a. AVG
- b. MIN
- c. All of the above
- d. SUM

Question 13

Complete

Mark 1.00 out of 1.00

Which key prevents insertion of a record that references a non-existent parent row?

- a. Foreign key
- b. Check constraint
- c. Primary key
- d. Unique key

Question 14

Complete

Mark 1.00 out of 1.00

Which key uniquely identifies a tuple in a table and cannot contain NULL values?

- a. Primary key
- b. Candidate key
- c. Superkey
- d. Foreign key

Question 15

Complete

Mark 1.00 out of 1.00

Which of the following best describes a composite key?

- a. A key ensuring referential integrity
- b. A key consisting of multiple attributes
- c. A key generated automatically by DBMS
- d. A key that cannot contain NULL values

Question 16

Complete

Mark 1.00 out of 1.00

Which of the following is a minimal superkey?

- a. Primary key
- b. Composite key
- c. Candidate key
- d. Foreign key

Question 17

Complete

Mark 1.00 out of 1.00

Which of the following is TRUE about candidate keys?

- a. Candidate keys depend on foreign key constraints
- b. All candidate keys must be chosen as primary keys
- c. A table can have multiple candidate keys
- d. Candidate keys can contain duplicate values

Question 18

Complete

Mark 1.00 out of 1.00

Which relationship allows an entity to be related to itself?

- a. One-to-one
- b. Many-to-many
- c. One-to-many
- d. Recursive relationship

Question 19

Complete

Mark 1.00 out of 1.00

Which SQL clause groups rows that have the same values into summary rows?

- a. GROUP BY
- b. ORDER BY
- c. DISTINCT
- d. HAVING

Question 20

Complete

Mark 1.00 out of 1.00

Which SQL clause is executed first in a SELECT query?

- a. GROUP BY
- b. SELECT
- c. FROM
- d. HAVING

Question 21

Complete

Mark 1.00 out of 1.00

Which SQL clause is used to filter aggregated results?

- a. HAVING
- b. FILTER
- c. GROUP BY
- d. WHERE

Question 22

Complete

Mark 1.00 out of 1.00

Which SQL query correctly finds departments having total salary > 1,00,000?

- a. SELECT dept, SUM(salary) FROM emp GROUP BY dept HAVING SUM(salary)>100000;
- b. SELECT dept, SUM(salary) FROM emp HAVING SUM(salary)>100000 GROUP BY dept;
- c. SELECT dept, SUM(salary) FROM emp GROUP BY dept WHERE SUM(salary)>100000;
- d. SELECT dept, SUM(salary) FROM emp WHERE SUM(salary)>100000 GROUP BY dept;

Question 23

Complete

Mark 1.00 out of 1.00

Which SQL query finds the number of employees per department?

- a. SELECT dept, COUNT(*) FROM emp HAVING dept GROUP BY dept;
- b. SELECT dept, COUNT(*) FROM emp GROUP BY dept;
- c. SELECT dept, COUNT(*) GROUP BY dept FROM emp;
- d. SELECT COUNT(*), dept FROM emp WHERE COUNT(*) GROUP BY dept;

Question 24

Complete

Mark 1.00 out of 1.00

Which SQL statement is valid for filtering rows before aggregation?

- a. WHERE salary > 50000
- b. WHERE AVG(salary) > 50000
- c. HAVING salary > 50000
- d. GROUP BY salary > 50000

Question 25

Complete

Mark 1.00 out of 1.00

Which clause is used to filter rows after aggregation?

- a. WHERE
- b. DISTINCT
- c. HAVING
- d. GROUP BY